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Beam Footprint Design, Scheduling, and Spectrum Assignment in Low Earth Orbit Mega-Constellations

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***Abstract*—Owing to the advantages of ubiquitous connectivity and high capacity, low earth orbit mega-constellations have drawn much attention. However, it is challenging to maximize network throughput under much available beam resource, complex inter-beam interference, and spatially uneven traffic demand with low computation complexity. In this paper, we formulate a novel beam management problem including beam footprint design, beam scheduling, and spectrum assignment. Considering the tight coupling among them, we first develop a fast beam footprint design method for earth-fixed cells, which adapts to dynamic satellite attitudes. Then, with the reformulation of objective and constraints, the primal problem is transformed into a beam scheduling and spectrum assignment problem with much smaller solution space, and an optimum algorithm is designed by combining branch-and-bound and interior point methods. Towards practical application, a more scalable graph-based heuristic algorithm is proposed. Simulation results show that the developed beam footprint design method can significantly reduce inter-beam interference and improve network throughput, and the proposed heuristic algorithm achieves comparable throughput but reduces execution time by two orders of magnitude compared with the optimum one. Moreover, in scenarios with 120 cells, our proposal increases network throughput by over 60% compared to the baseline based on integer recovery.**

***Index Terms*—Low earth orbit mega-constellation, beam footprint design, beam scheduling, spectrum assignment.**

I. Introduction

With the decreased manufacturing and launch costs as well as increased communication capacity per satellite, we have been witnessing explosive growth in the scale of low earth orbit (LEO) satellite networks [1]–[3]. Several famous companies like OneWeb, Amazon, and SpaceX have proposed mega-constellation plans with thousands of LEO satellites [4]. Particularly, SpaceX has operated more than 8000 LEO satellites to provide high data rate, which means that there are multiple visible satellites for a cell [5], [6]. Facing the challenges caused by complex inter-beam interference and uneven traffic distribution, effective beam management is the key for LEO mega-constellations [7]. In the following, related works are presented and summarized.

A. Inter-beam interference mitigation

There are several optional strategies to mitigate inter-beam interference, including turning off beams, precoding, spatial isolation, polarization isolation, and spectrum assignment. Literature [8] proposes a feasible strategy to turn off partial satellite beams to alleviate interference. Unfortunately, the decreased interference comes at the price of low beam resource utilization and network capacity. To maintain fairness, precoding techniques are adopted in [9] to satisfy the minimal rate constraint or maximize the minimal signal-to-interference-plus-noise ratio (SINR) among users, where semidefinite programming is leveraged to reduce computational complexity. Although precoding technique can mitigate inter-beam interference, their performance highly depends on accurate instantaneous channel state information [10], signal synchronization, and information coordination among satellites [11]. Therefore, spatial isolation, polarization isolation, and spectrum assignment are preferred for inter-beam interference mitigation in LEO mega-constellations.

Currently, most studies on spatial isolation in satellite networks focus on scenarios with full frequency reuse among beams [5], [7], [12]. Specifically, to alleviate inter-beam interference, they adopt an interference control method in which two adjacent cells obtain service in different time slots. In terms of polarization isolation, the authors in [13] propose a static scheme, where adjacent cells are configured with different antenna polarization modes. Nevertheless, when the number of active beams is large, static interference management schemes will cause severe inter-beam interference since the polarization configurations and orthogonal frequency bands are relatively scarce. To cope with this challenge, several researchers further combine spatial isolation with dynamic spectrum assignment [14]–[16]. In [14], a dynamic frequency reuse algorithm based on depth-first search is proposed. To enhance network capacity, authors in [15] devise a demand driven beam densification approach based on graph theory, which also supports dynamic frequency reusing. A flexible spectrum assignment strategy is developed in [16] by leveraging deep reinforcement learning, aiming to improve the SINR of cells.

B. Satellite beam scheduling facing uneven traffic demand

To tackle the challenges caused by uneven traffic demand distribution among cells, many studies pay attention to beam scheduling design. To fill the capacity-demand gap, the authors

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in [7] propose a joint beam scheduling and time slot assignment scheme with matching theory. In [17], an iterative beam serving time slot allocation algorithm is proposed to maximize user satisfaction. Although the above methods present high performance, they focus on scenarios with a single satellite, where inter-satellite interference are not taken into account. For scenarios with two satellites, authors in [12] propose a heuristic algorithm to ensure the transmission of high-prioritized service and maximize the ratio of capacity and traffic requests of cells. For multi-satellite scenarios, authors in [5] develop a beam scheduling method based on load balancing and inter-beam interference control, and simulation results show that the proposed method can improve traffic satisfaction rate and reduce unmet capacity.

C. Literature review summary and our contribution

Although the above studies obtain good performance in beam scheduling and interference mitigation, two deficiencies must be carefully handled for LEO mega-constellations. Firstly, current literature assumes inter-beam interference is nonexistent between nonadjacent cells. However, due to the high altitude of satellites and the spherical shape of the Earth, the beam footprint on the ground significantly deforms, which incurs severe inter-beam interference between nonadjacent cells. Specifically, according to [1], given the half-power beamwidth angle of 6° and the satellite altitude of 550 km, when the elevation angle between the cell center and satellite drops from 90° to 30°, the corresponding radius of the beam footprint increases 3.6 times. Therefore, the antennas model with fixed beamwidth in [4], [12], [18], [19] cannot fully unleash the network capacity of LEO mega-constellations.

To cope with this challenge, several methods are developed based on phased antenna array. The authors of [20] propose a beam footprint design method for a geostationary orbit satellite. However, due to the high speed and dynamic attitude of LEO satellites, the desired beam pattern for a satellite-cell pair is unknown and changes rapidly over time. Meanwhile, when cell shapes are regular hexagons, as the elevation angle between cells and satellites increases, the cell projection may split into two discontinuous parts or resemble rectangles in angle domain observation coordinates of LEO satellites, which causes significant energy leakage due to large errors in the elliptical fitting strategy. In [21], a non-iterative algorithm is proposed to calculate antenna weights based on priori based beam patterns. Unfortunately, the error between the desired footprint design and the obtained footprint pattern impacts user service quality and may cause inter-beam interference. Therefore, the above methods cannot be directly applied in LEO mega-constellations without modification.

Secondly, current studies focus on scenarios with limited beam resource and beam scheduling schemes are designed under static spectrum assignment strategies. When the number of active beams exceeds the number of cells, static spectrum assignment strategies will cause serious inter-beam interference and low spectral efficiency [22]. Since LEO mega-constellations feature much available beam resource, it is challenging to design a beam scheduling method with low computational complexity and high scalability.

To fill the research gap, this paper proposes an effective beam management framework for scenarios with LEO mega-constellations, earth-fixed cells, and spatially uneven traffic demand. Facing the challenges caused by beam deformation, an adaptive beam footprint design method is developed for earth-fixed cells, which can guarantee the targeted transmitting gain within cells and reduce interference to other cells. With such a method, the primal beam management problem is transformed into a more tractable beam scheduling and spectrum assignment problem based on the reformulation of objective as well as constraints, aiming to reduce the dimension of solution space. Subsequently, under inter-cell interference and available beam number constraints, an optimum beam scheduling and spectrum assignment algorithm based on branch-and-bound and interior point methods is developed to maximize network throughput. In addition, a low-complexity graph-based heuristic algorithm is further designed for practical application. The main contributions of this paper are summarized as follows.

- A novel communication network model with LEO mega-constellations is established, which features much available beam resource, phased antenna arrays at satellites, two antenna polarization modes, flexible spectrum assignment, inter-beam interference, earth-fixed cells, and spatially uneven traffic distributions. Based on this, we formulate a beam management problem consisting of beam footprint design, beam scheduling and spectrum assignment. Since the concerned problem is a complex mixed integer nonlinear programming and hence NP-hard, it is further decomposed into two subproblems, namely beam footprint design problem and the problem with beam scheduling and spectrum assignment.
- To overcome the interference between non-adjacent cells led by beam deformation, a fast beam footprint design scheme adapting to dynamic satellite attitude is proposed by leveraging cell projection fitting and non-iterative antenna weight calculation. Specifically, a coordinate transformation method is presented to obtain the area projection of earth-fixed cells in the local coordinate system of LEO satellites. Then, the beam footprint pattern with transition area is carefully designed to match the cell projection exactly, which provides a bridge to the advanced beamforming algorithms applied in LEO satellite networks.
- With the proposed beam footprint design method and the reformulation of objective as well as constraints, the primal problem is transformed into a more tractable beam scheduling and spectrum assignment problem, which significantly reduces the dimension of solution space. Then, a branch-and-bound algorithm combining interior point method is adopted to obtain the optimal solution. Towards practical implementation, a more scalable graph-based heuristic algorithm is developed, which can reduce computational complexity from exponential level to polynomial level.
- Extensive simulation results show that the adaptive beam footprint design method can improve network throughput and alleviate inter-beam interference between nonadja-

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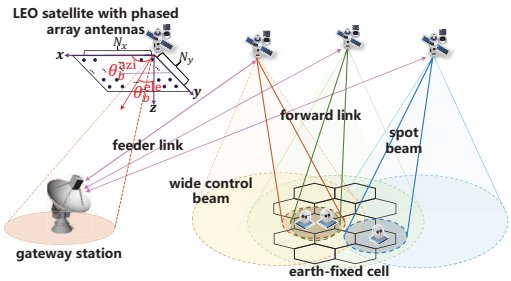

Fig. 1. The concerned scenario of LEO mega-constellations.

cent cells. In addition, the proposed graph-based heuristic algorithm can reduce the execution time by two orders of magnitude while achieving comparable throughput compared with the optimum one. Impressively, when satellites operate at 2 GHz and users are handheld terminals, our heuristic algorithm increases network throughput by over 60% compared with the baseline based on integer recovery in the scenario with 120 cells, 23 satellites and 184 active beams.

The reminder of this paper is organized as follows. Section II introduces the system model, and the concerned beam management problem with its solution is presented in Section III. In Section IV, the adaptive beam footprint design method is developed, and Section V provides optimum and suboptimal methods for beam scheduling and spectrum assignment decisions. Simulation results are illustrated in Section VI, and the conclusions are drawn in Section VII. The main notations used in this paper are summarized in Table I.

TABLE I
NOTATION

Notation	Definition
$\mathcal{S}$	The set of satellites
$\mathcal{B}_s$	The spot beam set of satellite s
M_s	The maximum number of spot beams of satellite s
$\mathcal{B}$	The set of all spot beams
$\mathcal{C}$	The set of cells
S	The number of visible satellites
C	The number of cells
B	The number of available beams
p_b	The polarization mode of beam b
Ω_b^1, Ω_b^2	The starting and ending frequencies of the operation band of beam b, respectively
W_b	The bandwidth of beam b
D_c	The downlink transmission demand of cell c
$\alpha_{c,b}$	The serving relationship between cell c and beam b
q	The sub-band index
$SINR_{c,b}^q$	The SINR of sub-band q when beam b serves cell c
$h_{c,b}$	The channel gain between the satellite generating beam b and cell c
$G_{c,b}$	The antenna gain on the direction from beam b to cell c
$\overline{n}$	Noise density
$INR_{c,b}$	The maximum INR of cell c among all sub-bands
N_x,N_y	The number of elements of phased antenna arrays on the x-axis and y-axis directions, respectively
$k_{m,n,b}$	The beamforming weight of the m-th row and the n-th column antenna element of beam b
$\theta_{c,b}^{\text{azi}}$, $\theta_{c,b}^{\text{ele}}$	The azimuth and the elevation angles of the cell c in the spherical observation coordinate system of beam b
$G_b(.)$	The antenna gain expression of beams
R_c	The throughput of cell c
$\overline{I}$	The inter-beam interference threshold
$\overline{B}$	The maximum number of serving beams of each cell
$\overline{W}$	System available bandwidth
$\overline{P}_{\text{antenna}}$	The antenna power
$\mathcal{A}_c$	The area of cell c on the ground
$\widetilde{\mathcal{A}}_{c,s}$	The 3 dB area of the desired beam footprint pattern
$\widehat{\mathcal{A}}_{c,s}$	The transition region of the desired beam footprint pattern
$\tilde{G}_{\text{antenna}}(.)$	The normalized desired beam footprint pattern
G_1 , G_2	The preset attenuation factors for $\tilde{G}_{\text{antenna}}(.)$
$\gamma_{c,s,p}$	The number of beams operating on polarization mode p that satellite s allocates to cell c
$W_{c,s,p}$	The beam bandwidth that satellite s allocates to cell c under polarization mode p
$\mathcal{B}_s^p$	The beam set of satellite s operating on polarization mode p
$R_{c,s,p}$	The transmission data rate that satellite s provides to cell c by polarization mode p
$P_{c,s}^{\text{beam}}$	The target power of the received signal of cell c from a beam generated by satellite s
$F_{b,b',c,c'}$	The variable to judge whether beams b and b' can share the spectrum when they serves cell c and c', respectively
$\overline{G}$	The receiving gain threshold
$\mathcal{V}$, $\mathcal{E}$	The set of vertices and edges in conflict graph
ν	The index of vertex in conflict graph
$e_{\nu,\nu'}$	The edge state between vertices ν and ν' in conflict graph
w_ν	The weight of vertex ν
$\tilde{W}_\nu$	The reallocated bandwidth of vertex ν
γ_ν	The beam scheduling decision $\gamma_{c,s,p}$ corresponding to vertex ν

II. System Model

A. Scenario description

As indicated in Fig. 1, we focus on resource management problem on the forward links of LEO mega-constellations. Define $\mathcal{S} = \{1, ..., s, ..., S\}$ as the set of LEO satellites. Assuming that satellites are equipped with uniform phased antenna arrays and can perform digital beamforming, which can adjust half-power beamwidth angles and the directions of beams flexibly [23]. Satellites simultaneously generate several wide control beams and spot beams, where the former is used to transmit control signaling, and the latter provides high data rate. Specifically, each point on the ground is simultaneously covered by multiple wide beams, and the control information, including the serving satellite, operating frequency, bandwidth, and the starting time and duration of service, is transmitted to users through wide control beams. In addition, wide control beams and spot beams operate on distinct spectrum bands, and consequently, inter-beam interference between them can be ignored. Meanwhile, we assume that LEO satellites perform frequency pre-compensation onboard, and then the Doppler effect is also ignored in the following. Considering that atmospheric drag perturbation, Earth shape irregularities, and orbital decay can cause deviations in satellite orbits and adjustments in their attitudes, advanced orbit error compensation [24] and precise attitude control mechanisms [25] are required to implement. With these approaches, we can optimistically assume that satellite antenna arrays remain oriented perpendicularly to the ground, thereby minimizing the adverse effects of orbital decay and atmospheric drag.

Denote the spot beam set of satellite s as $\mathcal{B}_s = \{b_{s,1}, b_{s,2}, ..., b_{s,M_s}\}$, where M_s is the maximum number of spot beams of satellite s. Define $\mathcal{B} = \cup_{s\in\mathcal{S}}\mathcal{B}_s$ as the set of all the available beams in the concerned scenario, and introduce variable b to indicate the beam index. Since the effect of dual polarization reuse is roughly equivalent to a two-color frequency reuse method without bandwidth loss [11], [13], several satellite constellations, including the Global Positioning System [26], the BeiDou System [27], and Starlink [28], have successfully implemented satellites with

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dual polarization modes. Therefore, two polarization modes are considered to mitigate inter-beam interference, e.g. left hand circular polarization (LHCP) and right hand circular polarization (RHCP). The polarization mode of beam b is denoted as $p_b \in \{1, 2\}$, and we assume that p_b is preset. To reduce inter-beam interference, a flexible spectrum assignment strategy is considered for beams, and define Ω_b^1 and Ω_b^2 as the starting and ending locations of the operation band of beam b in frequency domain, respectively. Denote the bandwidth of beam b as W_b, and $W_b = \Omega_b^2 - \Omega_b^1$.

Denote the set of earth-fixed cells as $\mathcal{C} = \{1, 2, ..., C\}$. Define D_c as the downlink transmission demands of cell c, which is the demand aggregated from all the users within cell c. Considering the uneven distribution of users, cells have differential transmission demands [5]. Meanwhile, a cell may simultaneously obtain service from multiple beams, which is an effective way of enhancing cell capacity in mega-constellation [1]. Moreover, due to limited hardware capabilities, each user only obtains service from a spot beam of a satellite, whereas different users in the same cell can be served by multiple satellites. Specifically, when multiple beams serve the same cell, sufficient polarization or frequency isolation among beams is required to alleviate inter-beam interference [11]. To characterize the state of beam b, a vector $(s_b, \alpha_{c,b}, \Omega_b^1, \Omega_b^2, W_b, p_b)$ is introduced, where s_b is the index of the satellite generating beam b and $\alpha_{c,b} \in \{0, 1\}$ is the serving relationship between beam b and cell c. Specifically, $\alpha_{c,b} = 1$ if beam b serves cell c, and $\alpha_{c,b} = 0$ otherwise.

Further, facing the challenges caused by complex inter-beam interference, long propagation delay, and heterogeneous traffic demand distribution, networks need to centrally determine beam operation states in advance [7]. The beam operation states are delivered to satellites via feeder links between them and the gateway station. In addition, since user-level time-frequency resource allocation has been comprehensively discussed in [29] and the interference among users can be omitted by allocating orthogonal channels [30], we only focus on cell-level spot beam management in this paper. Specifically, in order to simplify the analysis, we use a single virtual user located at the center of cells to replace randomly distributed users [5].

B. Transmission model

The duration of a single slot is set to a large value, and then the adverse impact of differential propagation delay between cells and satellites becomes nearly negligible. Hence, inter-beam asynchronous interference is ignored in the subsequent analysis. The spectrum of beam b is further divided the into N_b equal sub-bands and variable q is introduced to indicate sub-band index. When beam b serves cell c, the SINR of sub-band q can be expressed as [31]

$$SINR_{c,b}^q = \frac{h_{c,b} G_{c,b} G_{c,b}^{\text{user}} P_b}{W_b \overline{n} + N_b I_{c,b}^q}, \tag{1}$$

where $h_{c,b}$ is the channel gain between the satellite generating beam b and cell c, P_b is the transmission power of beam b. $G_{c,b}$ represents the antenna gain on the direction from beam b to cell c, $G_{c,b}^{\text{user}}$ is the receiving gain on the direction from users in cell c to the satellite generating beam b, $\overline{n}$ is noise density, and $I_{c,b}^q$ is the suffered interference strength of cell c on the sub-band q of beam b. Specifically, according to the satellite-to-ground pathloss model in [32], the channel gain $h_{c,b}$ can be calculated as

$$h_{c,b} = p_{c,b}^{\text{los}} h_{c,b}^{\text{los}} + (1 - p_{c,b}^{\text{los}}) h_{c,b}^{\text{nlos}}, \tag{2}$$

where $p_{c,b}^{\text{los}}$ is the line-of-sight probability of the channel between beam b and cell c, $h_{c,b}^{\text{los}}$ and $h_{c,b}^{\text{nlos}}$ are the channel gains of line-of-sight path and non-line-of-sight path, respectively. Moreover, $h_{c,b}^{\text{los}}$ and $h_{c,b}^{\text{nlos}}$ can be expressed as [18]

$$\begin{cases} h_{c,b}^{\text{los}} = PL_{c,b}^{\text{f}} + PL_{c,b}^{\text{g}} + CL_{c,b}^{\text{los}} \\ h_{c,b}^{\text{nlos}} = PL_{c,b}^{\text{f}} + PL_{c,b}^{\text{g}} + CL_{c,b}^{\text{nlos}}, \end{cases} \tag{3}$$

where $PL_{c,b}^{\text{f}}$ is the sum of free space path loss and shadow fading loss, $PL_{c,b}^{\text{g}}$ is the path loss caused by atmospheric gasses, and $CL_{c,b}^{\text{los}}$ and $CL_{c,b}^{\text{nlos}}$ are the clutter loss of line-of-sight path and non-line-of-sight path, respectively.

Define $\theta_{c,b}^{\text{azi}}$ and $\theta_{c,b}^{\text{ele}}$ as the azimuth and the elevation angles of the cell c in the spherical observation coordinate system of beam b, respectvely. According to [21], the antenna gain $G_{c,b}$ in (1) can be calculated as

$$\begin{aligned} G_{c,b} &= G_b(\theta_{c,b}^{\text{azi}}, \theta_{c,b}^{\text{ele}}, k_{m,n,b}) \\ &= \sum_{m=1}^{N_x} \sum_{n=1}^{N_y} k_{m,n,b} e^{-j \frac{2\pi}{\lambda} \vec{\xi}(\theta_{c,b}^{\text{azi}}, \theta_{c,b}^{\text{ele}}) \vec{\zeta}(m,n)}, \end{aligned} \tag{4}$$

where $k_{m,n,b}$ is the beamforming weight of the m-th row and the n-th column antenna element, N_x and N_y denote the number of elements of phased antenna arrays on the x-axis and y-axis directions, respectively. λ is the wavelength of transmitted signals, $\vec{\xi}(\theta_{c,b}^{\text{azi}}, \theta_{c,b}^{\text{ele}}) = (\cos\theta_{c,b}^{\text{azi}} \sin\theta_{c,b}^{\text{ele}}, \sin\theta_{c,b}^{\text{azi}} \cos\theta_{c,b}^{\text{ele}}, \cos\theta_{c,b}^{\text{ele}})$, and $\vec{\zeta}(m,n) = ((m-1)d, (n-1)d, 0)$ [23], and d is antenna element spacing. Based on the expressions (2)-(4), inter-beam interference strength $I_{c,b}^q$ in (1) can be calculated as [16]

$$I_{c,b}^q = \sum_{b' \in \mathcal{B}_b} P_{b'} G_{c,b'} h_{c,b'} \omega_{b,b',q} G_{\text{user}}(\theta_{b,c,b'}), \tag{5}$$

where $\mathcal{B}_b$ is the set of other beams with the same polarization mode as beam b, $P_{b'}$ is the transmission power of beam b', $h_{c,b'}$ is the channel gain between the satellite generating beam b' and cell c. $\omega_{b,b',q} = \frac{\Lambda_{b,b',q} W_b}{W_{b'} N_b}$, where $\Lambda_{b,b',q} \in \{0, 1\}$ represents whether beam b' occupies the sub-band q of beam b, W_b is the bandwidth of beam b. $G_{\text{user}}(\theta_{b,c,b'})$ is the receiving gain of users on the direction determined by off-axis angle $\theta_{b,c,b'}$, and $\theta_{b,c,b'}$ can be calculated based on the locations of the center of cell c and satellites generating beams b and b' [33].

When cell c is served by beam b, according to (5), the maximum interference-to-noise ratio (INR) of cell c among all sub-bands can be expressed as

$$INR_{c,b} = \max\left(\frac{I_{c,b}^q N_b}{W_b \overline{n}}\right), \forall q \in \{1, 2, ..., N_b\}, \tag{6}$$

where $\overline{n}$ is noise density. In addition, the throughput of cell c can be expressed as [5], [17]

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$$R_c = \min\{\sum_{b\in\mathcal{B}}\sum_{q=1}^{N_b}\frac{\alpha_{c,b}W_b}{N_b}\log(1+SINR_{c,b}^{q}), D_c\}, \quad (7)$$

where D_c is the downlink transmission demand of cell c.

III. Problem Formulation and Beam Management Framework

A. Problem constraints and formulation

First, define $\beta_{c,s} \in \{0,1\}$ representing whether the elevation angle between the center point of cell c and satellite s is above a preset threshold. Specifically, $\beta_{c,s} = 1$ if the satellite s is visible to cell c. Then, we have the following constraints

$$\alpha_{c,b} \le \beta_{c,s_b}, \forall c, \forall b \in \mathcal{B}, \quad (8)$$

where s_b is the index of the satellite generating beam b. The above constraints mean that the cell c can obtain services from the beams of the satellite s_b only if $\beta_{c,s_b} = 1$. In addition, a beam only serves up to one cell at a time, which can be expressed as

$$\sum_{c}\alpha_{c,b} \le 1, \forall b \in \mathcal{B}. \quad (9)$$

Inter-beam interference alleviation is essential to beam management. Here, the INR metric is applied to restrict inter-beam interference [12], which can be written as

$$\alpha_{c,b}I_{c,b} \le \overline{I}, \forall c \in \mathcal{C}, \forall b \in \mathcal{B}, \quad (10)$$

where $\overline{I}$ is the acceptable maximum INR threshold. Meanwhile, a lower INR corresponds to a higher SINR, which contributes to improving network throughput [34]. In addition, to guarantee fairness among cells, we limit the maximum number of serving beams allocated to a cell, which is given by

$$\sum_{b=1}^{B}\alpha_{c,b} \le \overline{B}, \forall c \in \mathcal{C}, \quad (11)$$

where $\overline{B}$ is a preset threshold. Moreover, define $\overline{W}$ as the whole available bandwidth of LEO mega-constellation. Then, the spectrum assignment constraints are expressed as

$$W_L \le \Omega_b^1 \le \Omega_b^2 \le W_L + \overline{W}, \forall b \in \mathcal{B}, \quad (12)$$

where W_L is the minimal frequency of the spectrum of LEO mega-constellation.

Assume that the total transmission power of each phased antenna array is $\overline{P}_{\text{antenna}}$, and hence we have [9], [35], [36]

$$\sum_{m=1}^{N_x}\sum_{n=1}^{N_y}|k_{m,n,b}|^2 = \overline{P}_{\text{antenna}}, \forall b \in \mathcal{B}, \quad (13)$$

where $\overline{P}_{\text{antenna}}$ is preset based on link budget and the number of beams it generates [18]. In addition, practical networks usually pursue the 3 dB contour of beam footprint to match the cell area exactly, which contributes to guarantee the quality of service within cells [23]. Considering that near-far distance effects are insignificant in LEO satellite networks [37], the differential path loss among users in the same cell is slight. Therefore, only the transmitting gain constraints are considered in this paper. Define $\mathcal{A}_c$ as the area of cell c on the ground. For each point $u_c \in \mathcal{A}_c$, define $(\theta_{u_c,b}^{\text{azi}}, \theta_{u_c,b}^{\text{ele}})$ as the direction from beam b to point u_c. Then, the constraints about the transmitting gain within the cell can be expressed as

$$\alpha_{c,b}\frac{G_b(\theta_{c,b}^{\text{azi}}, \theta_{c,b}^{\text{ele}}, k_{m,n,b})}{G_b(\theta_{u_c,b}^{\text{azi}}, \theta_{u_c,b}^{\text{ele}}, k_{m,n,b})} \le 2, \forall u_c \in \mathcal{A}_c. \quad (14)$$

With the above practical constraints, the beam management problem for LEO mega-constellations is formulated as follows.

$$\boldsymbol{P_0}: \max_{\{\alpha_{c,b},\Omega_b^1,\Omega_b^2,k_{m,n,b}\}}\sum_{c=1}^{C}R_c \quad (15)$$

$$s.t.\ \alpha_{c,b} \in \{0,1\}, \forall c \in \mathcal{C}, \forall b \in \mathcal{B}, \quad (15a)$$

$$(8)-(14).$$

The goal of problem $\boldsymbol{P_0}$ is to maximize network throughput.

B. Proposed beam management framework

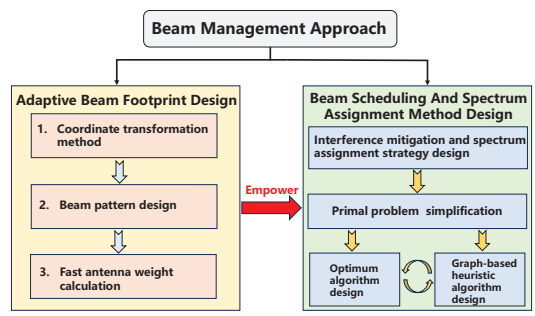

Fig. 2. The framework of the proposed beam management approach for LEO mega-constellations.

Note that the concerned problem involves both binary variables $\alpha_{c,b}$ and continuous variables Ω_b^1 as well as Ω_b^2. In addition, the objective function is the sum of fractional functions with binary variables, making it non-linear. Consequently, problem $\boldsymbol{P_0}$ is a complex mixed integer nonlinear programming problem, which has been proven to be NP-hard [9], [12]. Meanwhile, it can also be observed that the design of the beam management approach is significantly complicated by complex inter-beam interference, which is influenced by beam footprint design, beam scheduling, and spectrum assignment. Therefore, it is highly challenging to obtain the optimum solution directly.

To achieve an effective beam management with low computational complexity, we first propose an adaptive beam footprint pattern design scheme, which primarily alleviates inter-beam interference. With this scheme, we can simplify the complex inter-beam interference constraints, converting the primary problem into a linear one. To address the challenges caused by binary and continuous variables, we derive several propositions and leverage graph theory to efficiently determine beam scheduling and spectrum assignment. To this end, the problem $\boldsymbol{P_0}$ is decomposed into two more tractable subproblems, namely beam footprint design problem and problem with beam scheduling and spectrum assignment. Finally, the framework of the proposed beam management approach is described in Fig. 2, and we briefly summarize our proposal as follows.

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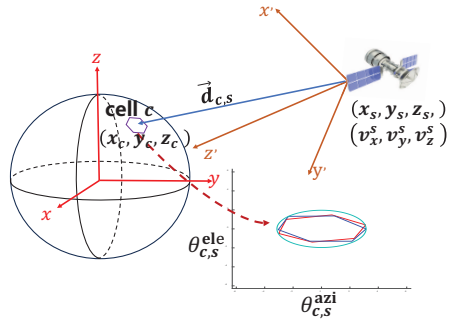

Fig. 3. An example of the proposed coordinate transformation method.

Note that the dynamic attitude of satellites and outdated channel state information are big obstacles to implement accurate precoding in practice [23]. Hence, it is more desirable to design a beam footprint pattern without explicit channel state information [21]. Firstly, we develop a coordinate transformation method, which can characterize the cell projection in the local coordinate systems of satellites with dynamic attitudes. Then, the 3 dB contour of desired beam footprint patterns is obtained by matching cell projection. Moreover, transition area of desired beam footprint patterns are designed to guarantee the targeted transmitting gain within cells taking the impact of Gibbs effect [21], [35] into account. Next, we quickly calculate antenna element weights by leveraging an efficient algorithm in [21]. The proposed adaptive beam footprint design method in Section IV can eliminate the negative effect of beam deformation phenomena, which contributes to alleviating inter-beam interference and enhancing network throughput due to less energy leakage.

In Section V, based on the proposed adaptive beam footprint design method, we further propose an easy-to-follow rule for LEO mega-constellations to judge whether two beams can share the same spectrum. Then, a more tractable throughput maximization problem is formulated, and a branch-and-bound algorithm combining interior point method is adopted to obtain optimal solution. Afterward, we also design a low-complexity graph-based heuristic algorithm for practical implementation.

IV. Adaptive Beam Footprint Pattern Design And Antenna Weight Calculation

A. Coordinate transformation

Note that the phased antenna array response in equation (4) depends on the relative position between satellites and cells. To adapt to different satellite attitudes, a unified coordinate transformation method is proposed. In the following, we first introduce this method based on a scenario as shown in Fig 3, where the normal line of phased array antenna passes through the center of the Earth. It is worth emphasizing that our proposal can easily be applied to other scenarios with different satellite attitudes.

Define (x_s, y_s, z_s) and (v_x^s, v_y^s, v_z^s) as the location and the velocity of satellite s in earth centered earth fixed (ECEF) coordinates, respectively. The x'- and z'-axis directions of local Cartesian coordinate system of satellite s are set to the movement direction of the satellite and the direction from the satellite to the Earth's center, respectively. Under the given velocity vector (v_x^s, v_y^s, v_z^s), we can obtain the unit vector of x'-axis based on the change of satellite position, and denote it as $\vec{e}_{x'} = (x_{s,1}, y_{s,1}, z_{s,1})$. Meanwhile, the unit vector of z'-axis is $\vec{e}_{z'} = (x_{s,3}, y_{s,3}, z_{s,3}) = (\frac{-x_s}{d_s}, \frac{-y_s}{d_s}, \frac{-z_s}{d_s})$, where $d_s = \sqrt{x_s^2 + y_s^2 + z_s^2}$. In addition, the y'-axis direction of the local Cartesian coordinate system of satellite s is determined with the right-hand rule, and its unit vector $\vec{e}_{y'} = (x_{s,2}, y_{s,2}, z_{s,2})$ can be calculated by vector product between $\vec{e}_{x'}$ and $\vec{e}_{z'}$, which is given by

$$\begin{cases} x_{s,2} = -z_{s,3}y_{s,1} + z_{s,1}y_{s,3} \\ y_{s,2} = -x_{s,3}z_{s,1} + x_{s,1}z_{s,3} \\ z_{s,2} = -y_{s,3}x_{s,1} + y_{s,1}x_{s,3}. \end{cases} \tag{16}$$

Define (x_c, y_c, z_c) as the center location of cell c in ECEF coordinates, and then the vector from satellite s to the center of cell c can be expressed as $\vec{d}_{c,s} = (x_c - x_s, y_c - y_s, z_c - z_s)$. Based on the obtained basis vectors, we can obtain the expression of vector $\vec{d}_{c,s}$ in the local Cartesian coordinate system of satellite s, which can be expressed as $\vec{d}_{c,s} = k_{c,s}^{x'}\vec{e}_{x'} + k_{c,s}^{y'}\vec{e}_{y'} + k_{c,s}^{z'}\vec{e}_{z'}$, where

$$\begin{bmatrix} k_{c,s}^{x'} \\ k_{c,s}^{y'} \\ k_{c,s}^{z'} \end{bmatrix} = \begin{bmatrix} x_{s,1} & y_{s,1} & z_{s,1} \\ x_{s,2} & y_{s,2} & z_{s,2} \\ x_{s,3} & y_{s,3} & z_{s,3} \end{bmatrix}^{-1} \begin{bmatrix} x_c - x_s \\ y_c - y_s \\ z_c - z_s \end{bmatrix}, \tag{17}$$

Subsequently, referring to [20], with the obtained Cartesian coordinates $(k_{c,s}^{x'}, k_{c,s}^{y'}, k_{c,s}^{z'})$, the center location of cell c in the local spherical coordinate system of satellite s can be expressed as $(|\vec{d}_{c,s}|, \theta_{c,s}^{\text{azi}}, \theta_{c,s}^{\text{ele}})$, where $\theta_{c,s}^{\text{azi}}$ and $\theta_{c,s}^{\text{ele}}$ represent the azimuth and elevation angles, respectively. Specifically, the location of satellite s is taken as the origin of the spherical coordinate system. Meanwhile, for each $b \in \mathcal{B}_s$, we have $\theta_{c,b}^{\text{azi}} = \theta_{c,s}^{\text{azi}}$ and $\theta_{c,b}^{\text{ele}} = \theta_{c,s}^{\text{ele}}$ in (4), where $\mathcal{B}_s$ is the spot beam set of satellite s.

B. Beam footprint design

Since the array response expression in (4) depends on azimuth and elevation angles, a reasonable beam footprint design in angular domain can avoid beam deformation [20]. When cell shapes are regular hexagons, as the elevation angle between cells and satellites increases, the cell projection may split into two discontinuous parts or resemble rectangles in the local angular observation domain of LEO satellites. In this case, relying on a single regular pattern for fitting cell projections results in substantial energy leakage due to significant errors. To address this challenge, a cell projection fitting algorithm based on distance estimation is proposed in Algorithm 1, and its details are elaborated as follows.

Define $\mathcal{L}_c = \{l_{c,1}, ..., l_{c,|\mathcal{L}_c|}\}$ as the location information vertex set of cell c in ECEF coordinates, where vertex coordinates are known prior. With the proposed coordinate transformation method, we can obtain the angle coordinates of each vertex $l \in \mathcal{L}_c$ in the local spherical coordinate system of satellites, which is expressed as $(\theta_{l,s}^{\text{azi}}, \theta_{l,s}^{\text{ele}})$. Then, for each point $(\theta^{\text{azi}}, \theta^{\text{ele}})$ in the local spherical coordinate system, the minimum distance between it and the vertices in $\mathcal{L}_c$ is

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Algorithm 1 Cell Projection Fitting Algorithm Based on Distance Estimation

1: Input: $\{\theta_{l,s}^{\text{azi}}, \theta_{l,s}^{\text{ele}}\}$, distance thresholds ψ_1 and ψ_2.
2: Calculate $d(\theta^{\text{azi}}, \theta^{\text{ele}})$ by (18) for all points in set $\mathcal{L}_c$.
3: Output: the half-power beamwidth contour $\widetilde{\mathcal{A}}_{c,s}$ and transition region $\widehat{\mathcal{A}}_{c,s}$ of the desired beam footprint pattern based on (19) and (20).

calculated, which is given by

$$d(\theta^{\text{azi}}, \theta^{\text{ele}}) = \min_{(\theta_{l,s}^{\text{azi}}, \theta_{l,s}^{\text{ele}})} (\theta^{\text{azi}} - \theta_{l,s}^{\text{azi}})^2 + (\theta^{\text{ele}} - \theta_{l,s}^{\text{ele}})^2. \quad (18)$$

Subsequently, the 3-dB area of beam footprint $\widetilde{\mathcal{A}}_{c,s}$ is obtained by comparing the minimum distance $d(\theta^{\text{azi}}, \theta^{\text{ele}})$ with preset threshold ψ_1, which can be expressed as

$$\widetilde{\mathcal{A}}_{c,s} = \{(\theta^{\text{azi}}, \theta^{\text{ele}}) | d(\theta^{\text{azi}}, \theta^{\text{ele}}) \leq \psi_1\}. \quad (19)$$

Due to the sharp transition band, the Gibbs effect causes a significant error between the obtained beam footprint pattern and the desired beam footprint pattern [21], [35], which cannot guarantee the targeted transmitting gain within cells. To smooth the negative influence of abruptly changed gain, we add a transition region in the beam footprint pattern. Denote the transition region as $\widehat{\mathcal{A}}_{c,s}$, and it is given by

$$\widehat{\mathcal{A}}_{c,s} = \{(\theta^{\text{azi}}, \theta^{\text{ele}}) | \psi_1 < d(\theta^{\text{azi}}, \theta^{\text{ele}}) \leq \psi_2\}, \quad (20)$$

where ψ_2 is a preset threshold.

Finally, the normalized desired beam footprint pattern for multi-beam antenna can be expressed as

$$\tilde{G}_{\text{antenna}}(\theta^{\text{azi}}, \theta^{\text{ele}}) = \begin{cases} 0 \text{ dB}, \ (\theta^{\text{azi}}, \theta^{\text{ele}}) \in \bigcup\limits_{c \in \mathcal{C}_s^{\text{antenna}}} \widetilde{\mathcal{A}}_{c,s}, \\ G_1 \text{ dB}, \ (\theta^{\text{azi}}, \theta^{\text{ele}}) \in \bigcup\limits_{c \in \mathcal{C}_s^{\text{antenna}}} \widehat{\mathcal{A}}_{c,s}, \\ G_2 \text{ dB}, \ \text{otherwise}, \end{cases} \quad (21)$$

where set $\mathcal{C}_s^{\text{antenna}}$ is the served cell set of the target antenna of satellite s, G_1 and G_2 are preset attenuation factor, respectively.

C. Fast antenna weight calculation

Inspired by recent advances in implementing efficient 2D-IFFT beamforming in satellites [38], we leverage a non-iterative algorithm in [21] to fast calculate antenna element weights. Specifically, the proposed algorithm in [21] bridges beam footprint design and filter windowing design, and the latter can utilize IFFT operations to accelerate calculation. To save space, we only briefly introduce the procedure of beam weight calculation and assume N_x and N_y are odd, where N_x and N_y are the number of antenna elements of phased antenna array on the x-axis and y-axis directions, respectively. Readers can refer to [21] for more details.

Define $\{f(m,n)\}$ as the tap weights of 2D finite impulse responses, and denote their magnitude response as $\{F(m',n')\}$ Moreover, $\{F(m',n')\}$ is uniformly sampled over the 2D-plane by using an $\eta N_x \times \eta N_y$ rectangular grid, where η is a constant and satisfies $1 \leq \eta \ll N_x$. Hence, we have $m, m' = 1, 2, ..., \eta N_x$, $n, n' = 1, 2, ..., \eta N_y$, $1 \leq \eta \ll N_x$. Firstly, we impose Hermitian symmetry on tap weights, which is given by

$$f(m,n) = f^*(\eta N_x + 1 - m, \eta N_y + 1 - n). \quad (22)$$

Algorithm 2 Adaptive Beam Footprint Design Algorithm

1: According elevation angle, fit cell projection by invoking Algorithm 1.
2: Obtain beam footprint pattern by (21).
3: Calculate antenna element weights $k_{m,n,b}$ by equations (22)-(26).
4: Output: $\{k_{m,n,b}\}$.

Define $\tilde{m} = \frac{2\pi m'}{\eta N_x} - \pi$ and $\tilde{n} = \frac{2\pi n'}{\eta N_y} - \pi$. Then, we establish the one-to-one mapping from $F(m',n')$ to normalized desired beam footprint pattern $\tilde{G}_{\text{antenna}}(\theta^{\text{azi}}, \theta^{\text{ele}})$, which is expressed as

$$F(m',n') = \begin{cases} 0, \ \sqrt{(\tilde{m})^2 + (\tilde{n})^2} > \pi, \\ \tilde{G}_{\text{antenna}}(\tilde{\theta}_1, \tilde{\theta}_2), \ \text{otherwise}, \end{cases} \quad (23)$$

where $\tilde{\theta}_1 = \arcsin(\sqrt{(\tilde{m})^2 + (\tilde{n})^2})$, and $\tilde{\theta}_2$ satisfies

$$\tilde{\theta}_2 = \begin{cases} \arctan(|\tilde{n}/\tilde{m}|), \ \tilde{m} \geq 0, \tilde{n} \geq 0, \\ \pi - \arctan(|\tilde{n}/\tilde{m}|), \ \tilde{m} \geq 0, \tilde{n} < 0, \\ \pi + \arctan(|\tilde{n}/\tilde{m}|), \ \tilde{m} \leq 0, \tilde{n} < 0, \\ 2\pi - \arctan(|\tilde{n}/\tilde{m}|), \ \tilde{m} < 0, \tilde{n} \geq 0. \end{cases} \quad (24)$$

Define $\tilde{F}(m',n') = F(m',n') e^{-j(\frac{\eta N_x + 1}{2} m' + \frac{\eta N_y + 1}{2} n')}$ and denote its IFFT result as $\tilde{f}(m,n)$. Then, the normalized antenna element weights $\tilde{k}_{m,n,b}$ can be calculated as

$$\tilde{k}_{m,n,b} = \begin{cases} \tilde{f}(m,n) e^{-j(m+n)\pi}, m \in [1, \frac{N_x+1}{2}], n \in [1, N_y], \\ (\tilde{k}_{N_x+1-m, N_y+1-n,b})^*, \text{otherwise}, \end{cases} \quad (25)$$

where $(.)^*$ represents conjugate operation. Subsequently, the antenna element weight $k_{m,n,b}$ in equation (4) is calculated as

$$k_{m,n,b} = \tilde{k}_{m,n,b} \times \sqrt{\frac{\overline{P}_{\text{antenna}}}{\sum_{m=1}^{N_x} \sum_{n=1}^{N_y} |\tilde{k}_{m,n,b}|^2}}, \quad (26)$$

where $\overline{P}_{\text{antenna}}$ is antenna power. Finally, the proposed adaptive beam footprint design method is summarized in Algorithm 2.

D. The application of our beam footprint design in practice

Due to the high-speed movement of LEO satellites, the designed beam footprint may violate transmitting gain constraints and result in severe inter-beam interference over time. Hence, the mobility of LEO satellites must be taken into consideration. Note that the proposed beam footprint design approach only relies on the relative positions of satellites and cells. With the availability of ephemeris information, the gateway station can pre-calculate the antenna weights based on our proposal for all potential serving relationships between satellites and cells in advance. Once the beam scheduling decision is made, the calculated antenna weights will be delivered to the corresponding satellites, which contributes to adapting to highly dynamic nature of LEO satellite networks.

In addition, when antennas adopt hybrid digital-analog antenna architecture, the antenna weights $\{k_{m,n,b}\}$ output by Algorithm 2 can be regarded as the optimal digital precoder. Then, the corresponding analog and digital precoders can be obtained based on the advanced algorithm, such as the algorithm presented in [39]. Hence, our proposal also can be extended to antennas with hybrid digital-analog antenna architectures.

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E. Complexity analysis

The proposed adaptive beam footprint design algorithm involves beam footprint pattern design and antenna weight calculation. According to [21], the computational complexity of antenna weight calculation can be expressed as $\mathcal{O}(2\eta^2 N_x^2 \log N_x)$, where η is a constant in (22) and N_x is the number of antenna element in x-axis. In addition, for our proposal based on distance estimation, calculating $F(m', n')$ in (23) requires obtaining the distance to each vertex in set $\mathcal{L}_c$. Therefore, the computational complexity of the beam footprint based on distance estimation can be expressed as $\mathcal{O}(2|\mathcal{L}_c|\eta^2 N_x^2 \log N_x)$, where $|\mathcal{L}_c|$ is the number of vertices in set $\mathcal{L}_c$.

V. Beam scheduling and spectrum assignment

A. Problem simplification

In the primal problem $\boldsymbol{P_0}$ of mega-constellations with much available beam resource, the number of variables and their feasible range are vast, which makes it challenging to design a beam management method with low complexity for practical implementation. In this part, with the proposed adaptive beam footprint design, we further transform the primal problem into a tractable form by reformulating the objective as well as constraints, and the details are presented as follows.

1) Objective function simplification: Due to identical channel loss and beamforming design, the strength of the received signals of different beams from a satellite is assumed to be the same at the center of a cell [16]. Introduce auxiliary variables $\gamma_{c,s,p}$ and $W_{c,s,p}$ representing the number of beams operating on polarization mode p and corresponding beam bandwidth that satellite s allocates to cell c. Specifically, we have $\gamma_{c,s,p} = \sum_{b' \in \mathcal{B}_s^p} \alpha_{c,b'}$, where $\mathcal{B}_s^p$ is the beam set of satellite s operating on polarization mode p, and $\alpha_{c,b'}$ represents the serving relationship between beam b' and cell c. When INR threshold is relatively small, the interference term can be ignored in (1). Then, the transmission data rate that satellite s provides to cell c under polarization mode p can be rewritten as

$$R_{c,s,p} = \sum_{b' \in \mathcal{B}_s^p} \alpha_{c,b'} W_{b'} \log(1 + \frac{P_{c,s}^{\text{beam}}}{W_{b'}\overline{n}}), \tag{27}$$

where $W_{b'}$ is the bandwidth of beam b', $P_{c,s}^{\text{beam}}$ is the target power of the received signal of cells from a beam generated by satellite s. Specifically, based on the proposed adaptive beam footprint design approach, power $P_{c,s}^{\text{beam}}$ can be assumed as a known parameter by leveraging advanced channel state precision methods [1], [40].

In addition, serving beams have distinct bands due to severe co-channel interference, and hence $\sum_{b' \in \mathcal{B}_s^p} \alpha_{c,b'} W_{b'} \leq W_{c,s,p}$. Then, we have the following proposition, whose proof is made in Appendix A.

Proposition 1. *When $W_{b'} = \frac{W_{c,s,p}}{\gamma_{c,s,p}}$ for $b' \in \mathcal{B}_s^p$ and $\alpha_{c,b'} = 1$, $R_{c,s,p}$ reaches the maximum.*

With **Proposition 1**, the maximum throughput can be reached when a satellite's beams with the same polarization mode are allocated to the same bandwidth to serve a cell, which can be expressed as

$$R_c^* = \min\{\sum_{s=1}^{S}\sum_{p=1}^{2} W_{c,s,p} \log(1 + \frac{\gamma_{c,s,p} P_{c,s}^{\text{beam}}}{W_{c,s,p}\overline{n}}), D_c\}, \tag{28}$$

where D_c is the transmission demand of cell c.

2) Interference mitigation and spectrum assignment simplification: Considering that polar isolation can prevent interference [11], we focus on interference among beams with the same polarization mode. Since the proposed adaptive beam footprint design has low sidelobe levels and can overcome the negative effect of beam deformation, severe inter-beam interference only appears in adjacent cells [5], [7]. Introduce $I_{b,b',c,c'}$ to indicate whether beams b and b' with the overlapping spectrum can simultaneously serve adjacent cells c and c', respectively. Specifically, $I_{b,b',c,c'} = 1$ if beams b and b' can share the spectrum, and $I_{b,b',c,c'} = 0$ otherwise. According to equation (5), the interference strength is also highly relative to the off-axis angle of receiving antennas. Therefore, we adopt off-axis angle based spectrum sharing model in [19] and [41] to judge whether two beams can share the same spectrum, which can be expressed as

$$I_{b,b',c,c'} = \begin{cases} 1, & G_{\text{user}}(\theta_{b',c,b}) < \overline{G} \ \& \ G_{\text{user}}(\theta_{b',c',b}) < \overline{G}, \\ 0, & \text{otherwise}, \end{cases} \tag{29}$$

where $\overline{G}$ is a preset threshold and readers can refer to [19] and [41] for more details. $\theta_{b',c,b}$ and $\theta_{b',c',b}$ are the off-axis angles formed by lines from the center of cells c and c' to satellite s and s', which generate beams b and b', respectively. The above expression means that beams with the same polarization mode can share the same spectrum to serve adjacent cells only if the receiving gain of interference signals is lower than a small threshold. Moreover, according to (29), $I_{b,b',c,c'}$ can be determined by the positions of satellites and cells, which contributes to simplifying interference analysis. In addition, when $I_{b,b',c,c'} = 1$ and satellites s_b and $s_{b'}$ allocate the overlapping spectrum to cells c and c', two positive value of beam scheduling decisions $\gamma_{c,s,p}$ and $\gamma_{c',s_{b'},p}$ will incur severe inter-beam interference, where s_b and $s_{b'}$ are the indices of the satellites generating beam b and b', respectively.

Introduce an auxiliary matrix $\boldsymbol{J}$ of dimension $2CS \times 2CS$, where the m-th row and m-th col correspond to variable $\gamma_{c,s,p}$ and $m = 2(c-1)s + 2(s-1) + p$. Particularly, $j_{m,n} = 1$ represents that severe interference occurs if corresponding beams share the same spectrum, and $j_{m,n} = 0$ otherwise. Denote the vector of all spectrum assignment decisions as $\boldsymbol{W}$, whose dimension is $2SC \times 1$. Specifically, the m-th row of $\boldsymbol{W}$ corresponds to decision $W_{c,s,p}$, where $m = 2(c-1)s + 2(s-1) + p$. Then, we can transform the interference constraints (10) and spectrum assignment constraints (12) into a more tractable form, which is given by

$$\boldsymbol{J}\boldsymbol{W} \leq \overline{\boldsymbol{W}}, \tag{30}$$

where $\overline{\boldsymbol{W}}$ is a vector with $2SC \times 1$ and the value of each element is the whole spectrum bandwidth $\overline{W}$.

3) Other constraint simplification: Noth that constraints (13) and (14) have been solved by the proposed adaptive beam footprint design method. Meanwhile, based on the above

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simplification processes, we have simplified the objective function and interference constraints by adopting $W_{c,s,p}$ and $\gamma_{c,s,p}$ to replace $\alpha_{c,s}$, Ω_b^1, and Ω_b^2. Next, we try to modify the remaining constraints to simplify the analysis. To achieve this goal, introduce an auxiliary variable Γ and $0 < \Gamma \ll \frac{1}{B}$ for constraint (8) , where B is the number of all available beams. Then, the visibility constraint (8) can be rewritten as

$$\Gamma\gamma_{c,s,p} \leq \beta_{c,s}, \forall c, \forall s, \forall p. \tag{31}$$

Moreover, constraints (9) can be replaced by the constraints of the available beam number of satellites, which is given by

$$\sum_{c=1}^{C}\sum_{p=1}^{2}\gamma_{c,s,p} \leq M_s, \forall s, \tag{32}$$

where M_s is the maximum number of spot beams of satellite s. In addition, constraint (11) limiting the serving beams for each cell can be rewritten as

$$\sum_{s=1}^{S}\sum_{p=1}^{2}\gamma_{c,s,p} \leq \overline{B}, \forall c. \tag{33}$$

4) Transformed problem: Based on the proposed adaptive beam footprint design approach and the above simplification transformations, the beam management problem can be converted into a more tractable beam scheduling and spectrum assignment problem as follows.

$$\begin{aligned} \boldsymbol{P_1}: &\max_{\{\gamma_{c,s,p}, W_{c,s,p}\}} \sum_{c=1}^{C} R_c \\ &s.t.\ (30)-(33). \end{aligned} \tag{34}$$

Particularly, compared with the primal problem under the given beam pattern design, the number of variables of the transformed problem is reduced from $3C(\sum_{s=1}^{S} B_s)$ to $4SC$, where B_s is the beam number of satellite s, which significantly reduces the dimension of solution space.

B. Optimum beam scheduling and spectrum assignment algorithm design

To tackle integer variables $\gamma_{c,s,p}$, we leverage the classical branch-and-bound method to design a beam scheduling and spectrum assignment algorithm, which can obtain the optimum solution to problem $\boldsymbol{P_1}$ [5]. Note that the branch-and-bound method requires relaxing integer variables $\gamma_{c,s,p}$ into continuous variables $\gamma'_{c,s,p}$ and then gradually closing to the optimum solution by adding constraints to limit the feasible range of $\gamma'_{c,s,p}$ [42]. Therefore, it is critical to adopt a low-complexity algorithm to obtain the optimum solution under arbitrary constraints of continuous variables $\gamma'_{c,s,p}$. To this end, we propose the following proposition, whose proof is made in Appendix B.

Proposition 2. *When variables $\gamma_{c,s,p}$ are continuous, problem $\boldsymbol{P_1}$ is a convex optimization problem.*

According to **Proposition 2**, the optimum solution to relaxed problems can be quickly obtained by traditional and efficient methodologies, e.g., interior point methods [44]. The adopted optimum beam scheduling and spectrum assignment algorithm is presented in Algorithm. 3. Specifically, we execute the following three steps to obtain an initial feasible solution [45].

Algorithm 3 Optimum Beam Scheduling And Spectrum Assignment Algorithm

Create an empty tree node list $\mathbb{T}$, and find a feasible solution as a node, which is then added to $\mathbb{T}$.
Initiate lower bound $\underline{R}_N = \underline{R}$.
while $\mathbb{T} \neq \emptyset$ **do**
  Take a node T off $\mathbb{T}$.
  if T is a feasible solution, and network throughput $R_T > \underline{R}_N$ **then**
    Record T, and $\underline{R}_N = R_T$.
  else
    Branch note T to produce two new nodes T_1 and T_2.
    Calculate the potential network throughput of nodes T_1 and T_2 by interior point methods and denote them as R_{T_1} and R_{T_2}, respectively.
    Add the obtained new node into the tree if potential network throughput is larger than $\underline{R}_N$.
  end if
end while
According to recorded node, output the optimum solution.

- 1. We relax all $\gamma_{c,s,p}$ and obtain the relaxed beam scheduling optimum decisions $\gamma'^{*}_{c,s,p}$ by interior point methods.
- 2. Round operations are iteratively performed on $\gamma'^{*}_{c,s,p}$ until all $\gamma'^{*}_{c,s,p}$ are recovered into integers. Remarkably, constraints (32) and (33) must hold in this process.
- 3. Under the fixed integer beam scheduling decisions, we rerun interior point methods to obtain spectrum assignment decisions and corresponding network throughput $\underline{R}$.

Under the obtained initial solution, we further perform branch-and-bound methods to obtain the optimum solution.

C. Graph-based heuristic algorithm

Note that branch-and-bound methods are computationally intensive, whose computational complexity significantly increases with the number of variables and constraint [42], [43]. Towards practical application, we further propose a graph-based heuristic algorithm with low complexity.

A conflict graph is first constructed to evaluate the benefits and negative effects of decisions $\gamma_{c,s,p}$ and $W_{c,s,p}$ [19]. Define $\mathcal{V}$ as the set of vertices in conflict graph, where the number of vertices is $2SC$, which ensures a one-to-one mapping between a vertex ν and a beam scheduling and spectrum assignment vector $(\gamma_{c,s,p}, W_{c,s,p}, c, s, p, P_{c,s}^{\text{beam}})$, as shown in Fig. 4. Denote the set of edges in conflict graph as $\mathcal{E}$, and if an edge connects two vertices ν and ν', it represents that the beam scheduling and spectrum assignment decisions relating to vertices ν and ν' will incur severe inter-beam interference. Introduce matrix $\boldsymbol{E}$ to characterize connection relationships among vertices, where $e_{\nu,\nu'=1}$ represents that vertex ν connects with vertex ν' in graph, and $e_{\nu,\nu'=0}$ otherwise. Moreover, $e_{\nu,\nu} = 1$ for all $\nu \in \mathcal{V}$. Here, define γ_ν and W_ν as the number of beams and total beam bandwidth decisions corresponds to vertex ν.

In Fig. 4, an example of a scenario with two cells and two satellites is presented, where each satellite is equipped with two beams operating on distinct polarization modes. Hence, the number of vertices in the conflict graph can be calculated as $2 \times 2 \times 2 = 8$, and all vertices are colored to represent different cells, satellites, and polarization modes. Moreover,

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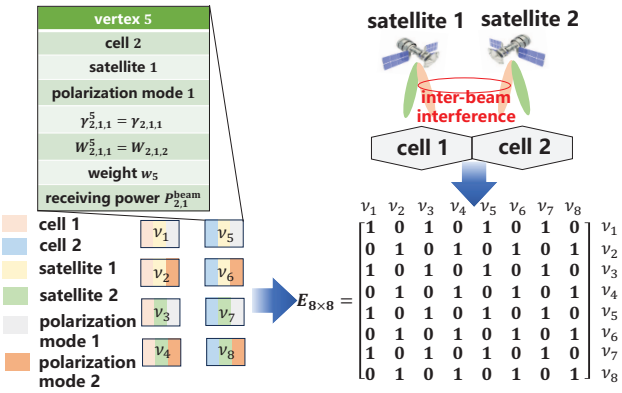

Fig. 4. An example scenario with two satellites and two cells and corresponding vertices and matrix $\boldsymbol{E}$ of conflict graph.

the edge matrix $\boldsymbol{E}$ is of size 8×8. Specifically, when two vertices are in conflict, the value of corresponding matrix element in $\boldsymbol{E}$ is set to 1. For example, vertices ν_2 and ν_8 are in conflict due to inter-beam interference, which corresponds to $e_{2,8} = e_{8,2} = 1$ in matrix $\boldsymbol{E}$. With such a graph, the proposed graph-based heuristic algorithm, which consists of three stages, is presented in Algorithm 4.

In the first stage, the network allocate beams to cells without incurring inter-beam interference, corresponding to unconnected vertices in the conflict graph. In this case, the maximum network throughput problem can be regarded as a maximum weighted independent set problem, where the weight of each vertex ν is set to

$$w_\nu = \overline{W} \log(1 + \frac{P^{\text{beam}}_{c,s}}{\overline{W}\overline{n}}), \tag{35}$$

where $\overline{W}$ is the bandwidth of all available spectrums, $P^{\text{beam}}_{c,s}$ is the received signal power from a beam relating to vertex ν. Define set $\mathcal{Z}$ as the maximum independent vertex set of the constructed conflict graph Next, a continuous greedy search procedure in [19] is adopted to obtain $\mathcal{Z}$. Specifically, in each step of the greedy search procedure, the optional vertex ν with the maximum value of $\frac{w_\nu}{\sum_{\nu' \in \mathcal{V}} e_{\nu,\nu'}}$ is selected, and corresponding beam scheduling and spectrum assignment decision are added to $\mathcal{Z}$, while all its adjacent vertices are marked as non-optional. Moreover, the greedy search procedure terminates when all vertices are either selected or non-optional. Then, we set corresponding $\gamma_\nu = 1$ and $W_\nu = \overline{W}$ for vertices in independent vertex set $\mathcal{Z}$. In stage 2, severe inter-beam interference must occur when any cell adds a serving beam. Therefore, we tackle this challenge through reasonable spectrum assignment strategy based on the following proposition, whose proof is made in Appendix C.

Proposition 3. *Assume that severe interference exists among B' beams, whose total available bandwidth is W', and the receiving signal strength from beam b is P_b. To maximize $\sum_{b=1}^{B'} W_b \log(1 + \frac{P_b}{W_b \overline{n}})$, the optimum spectrum assignment decisions of beams are $W_b = \frac{P_b W'}{\sum_{b=1}^{B'} P_b}$.*

In stage 2, beams are allocated to the cell one by one. At each step, the serving decision is determined by tradeoffing

Algorithm 4 Graph-based Heuristic Algorithm

Stage 1 :
Construct conflict graph, and initial $\mathcal{Z} = \emptyset$.
repeat
  Pick a vertex $\nu \in \mathcal{V}$ that maximize $\frac{w_\nu}{\sum_{\nu' \in \mathcal{V}} e_{\nu,\nu'}}$.
  $\mathcal{Z} = \mathcal{Z} \cup \{(\gamma_\nu, W_\nu)\}$, where $\gamma_\nu = 1$ and $W_\nu = \overline{W}$.
  Mark ν and its neighbors as non-optional.
until All vertices in $\mathcal{V}$ are non-optional
Stage 2 :
Calculate cell throughput R'_c based on $\mathcal{Z}$.
repeat
  Construct conflict subgraph, whose vertex set is denoted as $\mathcal{V}'$.
  Calculate the weight of all vertices based on (38).
  Pick a vertex $\nu \in \mathcal{V}'$ that maximizes $\frac{w'_\nu}{\sum_{\nu' \in \mathcal{V}'} e_{\nu,\nu'}}$.
  $\gamma_\nu = \gamma_\nu + 1$.
  Update spectrum assignment decisions and R'_c.
until No beams remain to be distributed
Stage 3 :
Under the given beam scheduling decisions of stage 2, obtain the final spectrum assignment decisions by interior point methods.
Output the beam management decisions and network throughput.

potential network throughput and negative effects on the subsequent decisions. First, we calculate the throughput of each cell and denote it as R'_c. Then, we construct a subgraph based on primal conflict graph. Define $\mathcal{V}'$ as the vertex set of the constructed subgraph. Particularly, vertex ν in primal conflict graph can remain in the subgraph only if corresponding beam scheduling decision $\gamma_\nu = 0$ and $\gamma_\nu = \gamma_\nu + 1$ meet constraints (32) and (33). Define $\mathcal{V}'_c$ as the set of vertices in the subgraph corresponding to cell c. To reduce computational complexity, we fix the maximum available bandwidth of each cell. Introduce auxiliary variable $\tilde{W}_\nu$ as the reallocated bandwidth of vertex ν when the value of γ_ν adds one and other beam scheduling decisions are fixed. According to **Proposition 3**, $\tilde{W}_\nu$ can be calculated as

$$\tilde{W}_\nu = \frac{P_\nu}{\sum_{\nu' \in \mathcal{V}'_c} \gamma_{\nu'} P_{\nu'}} W_c. \tag{36}$$

where $\gamma_{\nu'}$ is the beam scheduling decision corresponding to vertex ν', P_ν and $P_{\nu'}$ are the receiving powers relating to vertices ν and ν', respectively. $W_c = \sum_{\nu' \in \mathcal{V}'_c} W_{\nu'}$. Then, when $\gamma_\nu + 1$ is valid, the throughput that satellite s provides to cell c under polarization mode p can be coarsely calculated as

$$R'_{c,s,p} = \tilde{W}_\nu \log(1 + \frac{P_\nu}{\tilde{W}_\nu \overline{n}}). \tag{37}$$

Subsequently, we obtain corresponding network throughput R_ν if the value of γ_ν adds one. In this case, the weight of vertex ν is set to

$$w'_\nu = R_\nu - \sum_{c=1}^{C} R'_c. \tag{38}$$

Based on the constructed conflict graph, we only select one vertex that maximizes $\frac{w'_\nu}{\sum_{\nu' \in \mathcal{V}'} e_{\nu,\nu'}}$. Subsequently, we set $\gamma_\nu = \gamma_\nu + 1$ and update spectrum assignment decisions as well as R'_c. The above procedure is consistently executed until no beams remain to be distributed. To improve network throughput in stage 3, we leverage interior point methods to update all spectrum assignment decisions under the beam scheduling decisions provided by stage 2.

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D. Complexity analysis and the application in practice

Note that the optimal algorithm is developed based on the branch-and-bound method, whose computational complexity depends on the total number of nodes visited in the search tree $\mathbb{T}$ of the Algorithm 3 [46]. However, the number of visited nodes is a priori unknown due to varying network conditions. Therefore, we only provide the computational complexity of the optimum algorithm in the worst case, which occurs when the search tree is reduced to a chain. In this case, the computational complexity of optimum beam scheduling and spectrum assignment algorithm (**Algorithm 3**) can be expressed as $\mathcal{O}(2^{2SC})$ [5], where S and C are the numbers of satellites and cells, respectively.

The proposed graph-based heuristic algorithm consists of three stages. The computational complexity of stage 1 is $\mathcal{O}(4S^2C^2)$ [19]. In stage 2, there are up to B iterations, and each iteration takes $2SC(\log(2SC)+1)$ operations to update the subgraph and sort weights in the worst case, where B is the available beam number. Therefore, the computational complexity of stage 2 is $\mathcal{O}(2BSC(\log(2SC)+1))$. The computational complexity of stage 3 depends on interior point methods and can be expressed as $\mathcal{O}(2SC\log\frac{1}{\epsilon})$, where $\epsilon > 0$ is the pecision [7]. Overall, the computational complexity of the proposed graph-based heuristic algorithm (**Algorithm 4**) is $\mathcal{O}(4S^2C^2 + 2BSC(\log(2SC)+1) + 2SC\log\frac{1}{\epsilon})$.

In addition, due to the high-speed motion of LEO satellites, frequent inter-satellite handovers pose a significant challenge for LEO mega-constellations. In this case, inter-satellite handover control approach in [41] can first be applied to determine the serving relationships between cells and satellites. Subsequently, the concerned optimization problem $\boldsymbol{P_1}$ can incorporate additional constraint to further restrict the feasible area of variable $\gamma_{c,s,p}$. Then, our proposal can be implemented to generate beam scheduling and spectrum assignment decisions.

VI. Simulation Results and Analysis

A. Simulation parameters

In this section, simulations are conducted under S band and Ka band, respectively. Specifically, the operation frequencies of the former and the latter are 2 GHz and 20 GHz, respectively. Due to the lack of real satellite data, a custom LEO mega-constellation with 4800 satellites is considered, where satellites are evenly distributed in 80 orbit planes. The orbit height and inclination angle are 550 km and $53°$, respectively. Each satellite is equipped with eight beams sharing 100 MHz bandwidth [48]. Particularly, the polarization mode of 4 beams is LHCP and others are RHCP. The earth-fixed cell is regular hexagon, and cell radius is 43.4 km. Moreover, the center location of the concerned area is $(110.6°E, 26.67°N)$ and the traffic demands of cells are generated based on [47]. Network topology states are randomly generated by AGI Systems Tool Kit tools, which can provide the location information of LEO satellites. Based on the known satellite and cell locations, we then construct the edge matrix $\boldsymbol{E}$ of conflict graphs.

The channel parameters are determined based on 3GPP 38.811 [18]. In Ka band, we consider users to be very small aperture terminals, while handheld terminals are simulated in S band. The minimum visible elevation angle between cells and satellites is $40°$. The phased antenna array at each satellite has 100×100 antenna elements, and the spacing between adjacent antenna elements is half wavelength. The available antenna power is 38 dBm for each beam. In addition, we assume that the transmission power is adaptively adjusted according to the link budget, aiming to maintain a similar receiving power $P_{c,s}^{\text{beam}}$. In (19)-(22), parameters ψ_1, ψ_2, G_1, G_2, and η are set to 0.1, 0.8, -3, -30, and 10, respectively. Meanwhile, the maximum iteration number ρ of the cell projection fitting algorithm is set to 10. The inter-beam interference threshold $\overline{I}$ is set to -3 dB and the receiving gain threshold $\overline{G}$ in (29) is set to -15 dB. According to [49], for a mega-constellation such as the Amazon Kuiper, 73% of points on Earth have more than 4 satellites in view, and hence we assume that there are 3 satellites in view of each cell. Given that the satellites adopt two polarization modes, the maximum number of serving beams of each cell $\overline{B}$ is set to 6. Main parameters are summarized in Table II.

TABLE II
Simulation parameter setting

Parameter	Value
Satellite number / orbit plane number	4800 / 80
Polarization mode	LHCP, RHCP
The number of beams per satellite	8
Beam number ratio under two polarization modes	1:1
Orbit height / inclination angle	550 km / $53°$
Operation frequency	2 GHz, 20 GHz
System available bandwidth $\overline{W}$	100 MHz
The center location of the concerned area	$110.6°E, 26.67°N$
Cell radius	43.3 km
G_1 / G_2	-3 / -30
ψ_1 / ψ_2	0.1 / 0.8
η / ρ	10 / 10
$\overline{I}$ / $\overline{G}$ / $\overline{B}$	-3 dB / -15 dB / 6
Elevaton angle threshold	$40°$

B. The performance of the adaptive beam footprint design

In this part, we compare the performance of the proposed adaptive beam footprint design algorithm with baselines by Monte Carlo simulations. The simulation scenarios consist of 10000 users randomly distributed in the cell, and the locations and velocity of satellites are also randomly generated based on different elevation angle parameters. Meanwhile, transmission power is fixed for different elevation angles under the same operation frequency. In addition, we consider two types of antennas: phased antenna arrays (PAA) and reflector antennas (RFA). Moreover, four benchmark schemes are considered as follows.

- Elliptical beamwidth, PAA: Referring to [20], satellites adopt phased antenna arrays, and the cell projections in the local spherical coordinate system of LEO satellite are approximated by ellipses. Moreover, antenna weight is obtained by fast antenna weight calculation method detailed in Section IV.C.
- Circular beamwidth, PAA: Referring to [20], satellites adopt phased antenna arrays, and the cell projections in the local spherical coordinate system of LEO satellite

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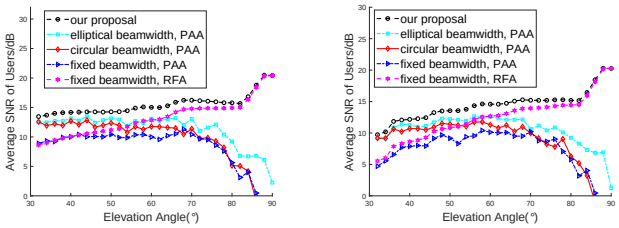
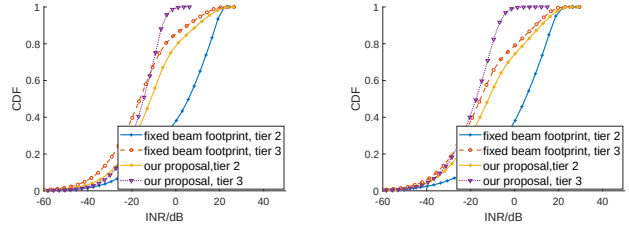


(a) (b)

Fig. 5. Figures present the performance comparison between our proposal and other benchmark schemes, where figure (a) is simulated under 2 GHz and figure (b) is simulated under 20 GHz.



(a) (b)

Fig. 6. Figures present the INR control performance of our proposal and the fixed beam footprint design based on PAA when elevation angle is $40°$, where figure (a) is simulated under 2 GHz and figure (b) is simulated under 20 GHz.

are approximated by circles. Moreover, antenna weight is obtained by fast antenna weight calculation method detailed in Section IV.C.

- Fixed beamwidth, PAA: Satellites adopt phased antenna arrays with fixed beamwidth [34]. Moreover, antenna weight is obtained by fast antenna weight calculation method detailed in Section IV.C.
- Fixed beamwidth, RFA: Beams are generated by typical reflector antennas with a fixed circular aperture, whose antenna pattern is provided in 3GPP 38.811 [18].

To verify the transmission performance of our proposal on downlink signal, SNR comparisons between our proposal and baselines are conducted, which is illustrated in Fig. 5. Compared to reflector antenna method with fixed beamwidth, our proposal based on distance estimation improves the average SNR as the elevation angle decreases. This phenomenon is because the projection area of cells in the local spherical coordinate system of LEO satellites decreases with the elevation angle, which contributes to concentrating the beam energy to serve users. Furthermore, under our proposal based on distance estimation, SNR generally increases with elevation angle, with only slight variations observed when the elevation angles are between $70°$ and $80°$, which is caused by the variation of pathloss. In contrast, it can be observed that the average SNR of beams with elliptical beamwidth shows obvious fluctuation as the elevation angle decreases. As mentioned in the Introduction, in cases of high elevation angles between cells and satellites, the cell projection may split into two discontinuous parts or resemble rectangles in the angle domain observation coordinates of LEO satellites. This causes a mismatch between the actual cell projection and the designed elliptical area, leading to significant energy leakage. Therefore, simulation results verify the superiority of our proposal. Specifically, when elevation angle is $40°$ and satellites operate at 20 GHz, our proposal achieves gains of approximately around 0.88 dB, 1.4 dB, 3.5 dB, and 4.5 dB compared to beams with elliptical beamwidth, beams with circular beamwidth, fixed beamwidth based on reflector antenna, and fixed beam pattern based on phase antenna array, respectively.

To demonstrate the capacity to overcome the interference between nonadjacent cells led by beam deformation, we evaluate the inter-beam interference mitigation performance of our proposal and the baseline. The simulation results are shown in Fig. 6, where the gain of receiving antennas on all directions is set to the maximum and the elevation angle between cells and satellites is $40°$. In simulation, cell layout have three tier, where tier 1 is the served cell, tier 2 represents the cells adjacent to the served cells, tier 3 represents nonadjacent cells, and the distance between served cell and other cells is less than four times cell radius. Remarkably, it can be seen that our proposal reduces interference compared with the fixed beam footprint design based on PAA in tier 3. Specifically, when adopting fixed beam footprint design based on PAA, the cumulative distribution functions (CDF) of INR show only 82% and 75.6% probability of being below -3 dB under 2 GHz and 20 GHz cases, respectively, resulting in severe inter-beam interference. In contrary, the CDFs of INR of our proposal have 97.4% and 98% probability of being below -3 dB under 2 GHz and 20 GHz cases, respectively. These results verify that our proposal can solve the challenges caused by beam deformation and the reasonable transformation assumption of interference mitigation constraint (29) in Section V-A. In addition, it can be observed that a crossing point appears between the fixed beam footprint and the proposed method in tier 3. This phenomenon can be explained as follows. When elevation angle is $40°$ and the half-power beamwidth angle is fixed, the radius of the beam projection increases by approximately 2.3 times compared with sub satellite point case. In this case, with fixed beam footprint schemes, some users are positioned near the first null of the antenna pattern, obtaining a low transmission gain. Thus, when the INR is low, the CDF value for the fixed beam pattern is higher than that of the proposed scheme. However, many users are still close to the 3 dB contour of the beam footprint and are affected by main lobe leakage. Therefore, a crossing point appears between the fixed beam footprint and proposed schemes in tier 3.

To assess the capacity of our proposal in guaranteeing the transmitting gain within cells, the obtained beam pattern is presented over the azimuth angular plane. The simulation result is presented in Fig. 7, where the baseline is the beam footprint design without a transition area $\widehat{\mathcal{A}}_{c,s}$ in (21), which ignore the negative effect of Gibbs effect. Two red dotted lines highlight the half-power beamwidth angle ranges of the desired beam pattern. If users are located at the cell edge, the baseline incurs around 8.7 dB gain loss compared with the cell center due to Gibbs effect. However, the difference in transmitting gain between the cell center and edge is 2.6 dB in our proposal,

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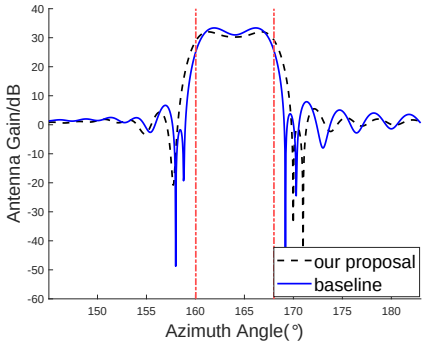
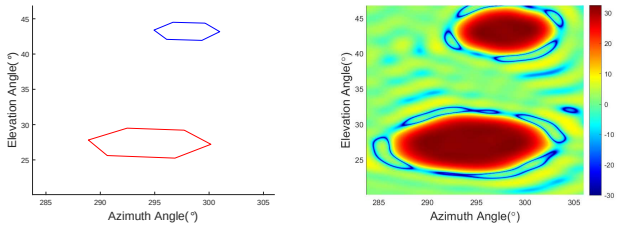


Fig. 7. The designed beam patterns of the proposed method and the baseline.



(a) (b)

Fig. 8. The performance of our proposal for multi-beam antennas, where figure (a) shows the cell projection of two target cells, and figure (b) presents the corresponding beam pattern.

which is very close to 3 dB. In addition, the performance of our proposed method for multi-beam antennas is shown in Fig. 8. Specifically, Fig. 8(a) illustrates the cell projections of two target cells and Fig. 8(b) presents the corresponding beam pattern. It can be observed that the obtained beam pattern accurately matches the projections of the two cells, which verifies the effectiveness of our proposal.

C. The performance of beam scheduling and spectrum assignment algorithm

In this section, we evaluate the performance of the proposed graph-based heuristic algorithm. Specifically, the downlink transmission demands for all cells are randomly generated based on the potential sum capacity of all visible satellites. Consequently, when the number of visible satellites is the same, the total transmission demands across different scenarios remain approximately equal in the following simulations.

First, we compare it with the optimum algorithm under operation frequency 2 GHz and 20 GHz, as shown in Fig. 9 and Fig. 10, respectively. Since the optimum algorithm is time-consuming, we only consider scenarios with a few cells. Moreover, there are five satellites visible to the concerned area. In addition, bars in figures represent network throughput, and lines indicate algorithm execution time. It can be observed that the developed graph-based heuristic algorithm achieves comparable network throughput with the optimum algorithm. Particularly, when the cell number is 7, the network throughput of the graph-based heuristic algorithm achieves more than 97% of the optimum throughput. Meanwhile, as shown in Fig. 10, the execution time of the optimal algorithm does not always

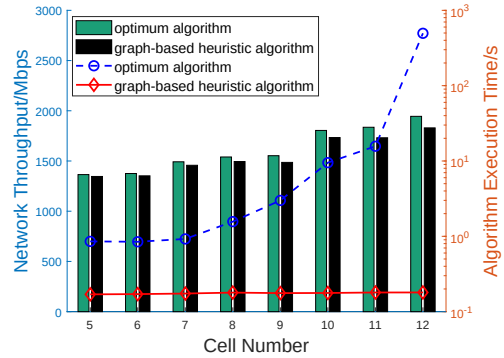


Fig. 9. The performance of the graph-based heuristic algorithm compared with the optimum algorithm in 2 GHz cases, where bars represent network throughput, and lines indicate algorithm execution time.

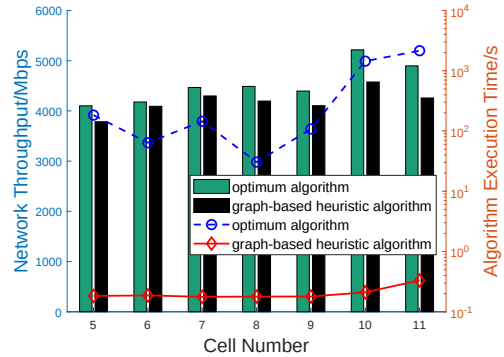


Fig. 10. The performance of the graph-based heuristic algorithm compared with the optimum algorithm in 20 GHz cases, where bars represent network throughput, and lines indicate algorithm execution time.

increase with the number of cells, which may be due to a reduction in the number of visited nodes. Nevertheless, the execution time of the optimal algorithm may fluctuate with the number of cells, but generally follows an exponentially increasing trend. Moreover, computational complexity of graph-based heuristic algorithm only slightly increases, which reduces the execution time by two orders of magnitude.

To evaluate the scalability of graph-based heuristic algorithm, the number of cells is further increased, which also corresponds to the increase in the number of visible satellites. Meanwhile, we consider five benchmark schemes as follows.

- Full frequency reuse [12]: All beams share the same spectrum without frequency division.
- Two color frequency reuse [13]: The spectrum is equally divided into two sub-bands, and each beam only occupies one sub-band.
- Greedy scheme [5]: Beams are allocated to the cell one by one. At each step, the network throughput is greedily maximized.
- Integer recovery [45]: The final beam scheduling decisions $\gamma_{c,s,p}$ are obtained by rounding the obtained non-integer solutions $\gamma'^{,*}_{c,s,p}$ of the first step in optimum algorithm into the nearest integers. Then, the spectrum assignment is determined by interior point methods.
- SCA-based [50]: In this method, the integer variables $\alpha_{c,b}$

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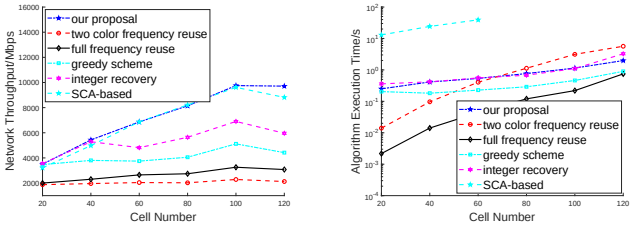
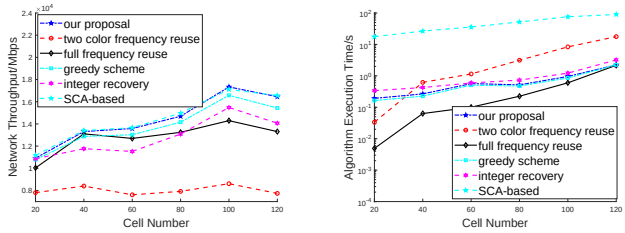
are first treated as continuous, and an additional penalty term is added to the objective function, which contributes to deriving a quasi-integer solution. Then, SCA technique is leveraged to construct convex functions to approximate the solution of the primary problem $\boldsymbol{P_0}$.



(a) (b)

Fig. 11. Figure (a) shows the network throughput of our proposal and baselines with operation frequency 2 GHz, and figure (b) illustrates their execution times.



(a) (b)

Fig. 12. Figure (a) shows the network throughput of our proposal and baselines with operation frequency 20 GHz, and figure (b) illustrates their execution times.

TABLE III
THE NUMBER OF VISIBLE SATELLITE

Cell number	20	40	60	80	100	120
Satllite number	11	14	16	19	23	23

The simulation results of network throughput and execution time under operation frequencies 2 GHz and 20 GHz are presented in Fig. 11 and Fig. 12, respectively. Table III illustrates the number of visible satellites. The results demonstrate the superiority of our proposal by comparing with baselines. The throughput performance of our method is comparable to that of the SCA algorithm, while showing a significant improvement over other comparison schemes. Specifically, when the number of cells is 120, the network throughput is increased by over 60% and 6% under operation frequency 2 GHz and 20 GHz, respectively. The reasons for the observed differences in two frequency bands are summarized as follows. Due to the lack of spatial isolation provided by receiving antennas, the negative effect of inter-beam interference is more severe for handheld terminals. Hence, it is more urgent for scenarios with handheld terminals to develop an effective beam management approach to control inter-beam interference. This conclusion can be verified by comparing the network throughput under 2 GHz and 20 GHz. In addition, although network throughput generally increases with the number of cells, several peak points and crossing points can be observed in Figs. 10, 11 and 12. The reason for this phenomenon can be summarized as follows. In our simulation scenarios, the serving duration of each beam is fixed, and the average transmission demand per cell decreases as the number of visible satellites remains constant. In this case, while an increase in the number of cells tends to create more relaxed conditions for beam management, the reduced average transmission demand per cell may negatively impact overall network throughput.

When the number of cells is 120, the number of visible satellites is 23, and the number of active beams is 184, the execution time of our proposal is only around 1-2 seconds, which is comparable to benchmark schemes, except for the SCA-based scheme, demonstrating the high scalability of our proposal. Moreover, the execution time of the SCA-based scheme is nearly 90 seconds. Therefore, although our proposed method achieves comparable network throughput to the SCA-based scheme, it significantly reduces the execution time, making it more suitable for LEO mega-constellations. Moreover, due to long propagation delay and massive satellites, it is appropriate for LEO mega-constellations to adopt a hybrid centralized-distributed beam management approach. To reduce algorithm execution time, all cells can be divided into multiple orthogonal groups, with only adjacent cells within the same groups. Then, cells that are adjacent to other groups are only allocated to beams with fixed polarization mode to mitigate inter-group interference. In this case, the beam management decisions of each group can be independently formulated based on our proposal, which significantly reduces the algorithm execution time for LEO mega-constellations.

VII. CONCLUSIONS

This article has studied beam management for LEO mega-constellations with large number of active beams and cells. Facing the challenges caused by beam deformation and tight coupling of beam scheduling, spectrum management, and beam footprint design, we have first developed a cell-level adaptive beam footprint design method for earth-fixed cells, consisting of coordinate transformation, desired beam pattern design, and fast antenna weight calculation, which can simplify interference analysis and guarantees stable received signal power with low computational complexity. Based on cell-level beam footprint design, we further transform the primal problem into a tractable beam scheduling and spectrum assignment problem by reformulating the objective as well as constraints, which contributes to significantly reducing the dimension of solution space. Then, a branch-and-bound method combining interior point method is adopted to obtain optimal beam scheduling and spectrum assignment decisions. Towards practical application, a more scalable graph-based heuristic algorithm has been developed. Extensive simulation results have shown that the proposed adaptive beam footprint design can enhance the receiving signal strength within the target cell and greatly reduced interference between nonadjacent cells. Moreover, the simulation results have demonstrated that the proposed graph-based heuristic algorithm can achieve near-optimal network throughput but with low complexity.

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Appendix A
Proof of Proposition 1

It can be observed that $R_{c,s,p}$ in (27) is a monotonously increasing function with regard to beam bandwidth. Therefore, $R_{c,s,p}$ reaches the maximum value only if $\sum_{b'\in\mathcal{B}_s^p}\alpha_{c,b'}W_{b'} = W_{c,s,p}$. Next, we prove **Proposition 1** by contradiction. Assume there exists i and j with $W_i < W_j$, $i,j \in \mathcal{B}_s^p$ and $\alpha_{c,i} = \alpha_{c,j} = 1$, which maximizes throughput. We fix spectrum assignment decisions of other beams and only focus on the scenario with two beams i and j. Introduce auxiliary variable $W_{i,j} = W_i + W_j$. The sum of throughput provided by beam i and j can be written as

$$\begin{aligned} R_{c,s,p}(W_i) = {} & W_i \log(1 + \frac{P_{c,s}^{\text{beam}}}{W_i\overline{n}}) \\ & + (W_{i,j} - W_i)\log(1 + \frac{P_{c,s}^{\text{beam}}}{(W_{i,j} - W_i)\overline{n}}). \end{aligned} \tag{39}$$

It can be observed that $R_{c,s,p}(W_i)$ is concave due to $\frac{\partial^2 R_{c,s,p}(W_i)}{\partial W_i^2} \leq 0$. Meanwhile, $R_{c,s,p}(W_i)$ has only one extreme point $W_i = \frac{W_{i,j}}{2}$, which is obtained by setting $\frac{\partial R_{c,s,p}(W_i)}{\partial W_i} = 0$. Therefore, $R_{c,s,p}(W_i) < R_{c,s,p}(\frac{W_{i,j}}{2})$ due to $W_i < W_j$. The above analysis contradicts the primary maximum throughput assumption, and hence it can be proven that **Proposition 1** always holds.

Appendix B
Proof of Proposition 2

According to **Proposition 1**, the data rate that a satellite provides to a cell by polarization mode p is calculated as

$$R(\gamma'_{c,s,p}, W_{c,s,p}) = W_{c,s,p}\log(1 + \frac{\gamma'_{c,s,p}P_{c,s}^{\text{beam}}}{W_{c,s,p}\overline{n}}). \tag{40}$$

Due to $R(\gamma'_{c,s,p}, W_{c,s,p} = 0) = 0$, $R(\gamma'_{c,s,p}, W_{c,s,p})$ is continuous in the domains of $\gamma'_{c,s,p}$ and $W_{c,s,p}$. Then, we calculate the second-order mixed derivatives of $R(\gamma'_{c,s,p}, W_{c,s,p})$, which can be expressed as

$$\begin{cases} \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial {\gamma'_{c,s,p}}^2} = -\frac{W_{c,s,p}{P_{c,s}^{\text{beam}}}^2}{\ln 2(W_{c,s,p}\overline{n}+\gamma'_{c,s,p}P_{c,s}^{\text{beam}})^2}, \\ \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial W_{c,s,p}^2} = -\frac{\gamma'_{c,s,p}P_{c,s}^{\text{beam}}\overline{n}(\gamma'_{c,s,p}P_{c,s}^{\text{beam}}\overline{n}W_{c,s,p})}{\ln 2(W_{c,s,p}\overline{n}+\gamma'_{c,s,p}P_{c,s}^{\text{beam}})^2(W_{c,s,p}\overline{n})^2}, \\ \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial W_{c,s,p}\partial\gamma'_{c,s,p}} = -\frac{\gamma'_{c,s,p}{P_{c,s}^{\text{beam}}}^2}{\ln 2(W_{c,s,p}\overline{n}+\gamma'_{c,s,p}P_{c,s}^{\text{beam}})^2}, \\ \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial\gamma'_{c,s,p}\partial W_{c,s,p}} = -\frac{\gamma'_{c,s,p}{P_{c,s}^{\text{beam}}}^2}{\ln 2(W_{c,s,p}\overline{n}+\gamma'_{c,s,p}P_{c,s}^{\text{beam}})^2}. \end{cases} \tag{41}$$

Next, we can obtain Hessian Matrix of $R(\gamma'_{c,s,p}, W_{c,s,p})$, which is given by

$$\boldsymbol{H_f} = \begin{bmatrix} \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial {\gamma'_{c,s,p}}^2} & \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial W_{c,s,p}\partial\gamma'_{c,s,p}} \\ \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial\gamma'_{c,s,p}\partial W_{c,s,p}} & \frac{\partial^2 R(\gamma'_{c,s,p},W_{c,s,p})}{\partial W_{c,s,p}^2} \end{bmatrix}. \tag{42}$$

Since $|\boldsymbol{H_f}| = 0$ and each element value in Hessian Matrix is negative, $R(\gamma'_{c,s,p}, W_{c,s,p})$ is negative semi-definite, which verifies that $R(\gamma'_{c,s,p}, W_{c,s,p})$ is concave. In this case, the objective of problem $\boldsymbol{P_1}$ is to maximize a concave function . Meanwhile, all constraints also are convex. Hence, problem $\boldsymbol{P_1}$ is a convex problem, and its local optimum solution is the global optimum solution.

Appendix C
Proof of Proposition 3

According to **Proposition 2**, the throughput expression is a concave function and only has one optimum point. By taking the partial derivative as 0, we can prove that the optimum bandwidth allocation decision is $W_b = \frac{P_b W'}{\sum_{b=1}^{B'} P_b}$.

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